

matrix

↳ Array of nos. = $a \rightarrow \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

$$a = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

$$D = \begin{bmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{bmatrix}_{i \times j}$$

$$i = 3$$

$$j = 3$$

$$D = d_{ij}^o$$

$i \rightarrow$ no. of rows

$j \rightarrow$ no. of columns

elements / coefficients / entries

$$a = \begin{bmatrix} 0 & 1 & 2 & 4 \\ 1 & -2 & 3 & 0.7 \\ 1 & 1 & 0 & 3 \end{bmatrix}_{3 \times 4}$$

3 rows 4 columns

vector

↓ 1 column

$$\vec{a} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{bmatrix}_{n \times 1} \text{ column matrix}$$

stacked | Block

$$A = \begin{bmatrix} B & C \\ D & E \end{bmatrix}$$

for each block
column must have
same no. of columns

$$\checkmark B = \begin{bmatrix} 0 & 2 & 3 \end{bmatrix}_{1 \times 3}$$

$$C = \begin{bmatrix} -1 \end{bmatrix}_{1 \times 1}$$

$$\checkmark D = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 5 \end{bmatrix}_{2 \times 3}$$

$$E = \begin{bmatrix} 4 \\ 4 \end{bmatrix}_{2 \times 1}$$

$$A = \begin{bmatrix} 0 & 2 & 3 & -1 \\ 2 & 2 & 1 & 4 \\ 1 & 3 & 5 & 4 \end{bmatrix}_{3 \times 4}$$

$$A = \begin{bmatrix} 0 & 2 & 3 & -1 \\ 2 & 2 & 1 & 4 \\ 1 & 3 & 5 & 4 \end{bmatrix} \times$$

$$E = \begin{bmatrix} 4 & 1 \\ 4 & 0 \end{bmatrix}$$

Special matrices

1.) Zero matrix $a = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \text{Square}$
 2×2

all the elements

$$b = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ } \underline{2 \times 3} \times$$

$$c = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ Square}$$

$\underline{3 \times 3}$

2.) Square matrix $\rightarrow \text{rows} = \text{columns}$

$$i = j \quad \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \text{ } 2 \times 2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \text{ } \underline{\text{non-zero}}$$

3. Identity matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = A_{ij}$$

$$\text{for } i = j \rightarrow A_{ij} = 1$$

$$\text{for } i \neq j \rightarrow A_{ij} = 0$$

4. Diagonal matrix

$$\text{for } i = j \rightarrow A_{ij} \neq 0$$

$$\text{for } i \neq j \rightarrow A_{ij} = 0$$

Sparse matrix

$A_{ij} = 0$ for majority of elements

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

$A + B =$ all the elements

one added $A_{ij} + B_{ij}$

$$A - B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1-1 & 0-1 \\ 0-2 & 1-3 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -2 & -2 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1+1 & 0+1 \\ 0+2 & 1+3 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}$$

Common operations on matrix

$$\rightarrow A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad 2 \times 2$$

$$\rightarrow B = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \quad 2 \times 2$$

$A + B$

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 1 \end{bmatrix} \quad 2 \times 3$$

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad 2 \times 3$$

$A + B$ ✓

$$\begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2+1 & 3+0 & 4+1 \\ 1+1 & 0+1 & 1+2 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 5 \\ 2 & 1 & 3 \end{bmatrix}$$

multiplication by scalar

$$\alpha A \rightarrow \alpha A_{ij}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \quad 3 \cdot A$$
$$3A = 3 \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & -3 \end{bmatrix}$$

Some properties

$$\rightarrow A + B = B + A$$
$$\rightarrow \alpha(A + B) = \alpha A + \alpha B$$
$$\rightarrow (A + B)^T = A^T + B^T$$

Transpose

$$(A)^T = (A_{ji}) \quad A_{ji}^T$$

e.g. $A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \\ 3 & 1 \end{bmatrix}$ $A^T = \begin{bmatrix} 0 & 2 & 3 \\ 1 & 0 & 1 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 3 \\ 4 & 1 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 1 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

$$(A^T)^T = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 3 \\ 4 & 1 & 1 \end{bmatrix}$$

$$(A^T)^T = A$$

matrix multiplication

$$A = \begin{bmatrix} A_1 & A_2 & A_3 \\ A_4 & A_5 & A_6 \\ A_7 & A_8 & A_9 \\ A_{10} & A_{11} & A_{12} \end{bmatrix}_{4 \times 3}$$

$$B = \begin{bmatrix} B_{13} & B_{14} \\ B_{15} & B_{16} \\ B_{17} & B_{18} \end{bmatrix}_{3 \times 2}$$

$$A \cdot B = X \quad X_{11}$$

$$C = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}_{2 \times 2}$$

$$A \cdot C \times$$

$$B \cdot C \checkmark$$

A · B

no. of columns A = no. of rows B

$$\underline{\underline{\vec{a} \cdot \vec{b} = c \checkmark}}$$

$$C_{ij} = a_{i1} b_{1j} + a_{i2} b_{2j} + \dots$$

$$= \sum_{k=1}^n a_{ik} b_{kj}$$

$$A \quad i \times j$$

$$B \quad j \times k$$

$$(A_{i \times j}) \cdot (B_{j \times k}) = \underline{\underline{C_{i \times k}}}$$

$$A = \begin{bmatrix} 2 & -2 \\ 5 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} -1 & 4 \\ 7 & -6 \end{bmatrix}$$

$$A \cdot B = C = \begin{bmatrix} -16 & 20 \\ 16 & 2 \end{bmatrix}$$

$$A \cdot B = \begin{bmatrix} 2 & -2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} -1 & 4 \\ 7 & -6 \end{bmatrix} = \begin{bmatrix} (2)(-1) + (-2)(7) & (2)(4) + (-2)(-6) \\ (5)(-1) + (3)(7) & (5)(4) + (3)(-6) \end{bmatrix}$$

$$C = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}_{2 \times 3} \quad \begin{bmatrix} u & v & w \\ w & x & y \\ y & z & z \end{bmatrix}_{3 \times 2}$$

$A \quad B$

$$C_{2 \times 2} = \begin{bmatrix} au + bw + cy & av + bx + cz \\ du + ew + fy & dv + ex + fz \end{bmatrix}_{2 \times 2}$$

$$\checkmark A \cdot B \neq B \cdot A$$

$$A(BC) = (AB)C$$

$$A(B+C) = AB + AC$$

$$(A+B)C = AC + BC$$

Special matrices

1) minor (square matrices)

$$A = \begin{bmatrix} \cancel{a_{11}} & \cancel{a_{12}} & \cancel{a_{13}} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

minor of a_{11} = $\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

2) Cofactor $\rightarrow a_{ij}^o = (-1)^{i+j} \times \text{minor of } a_{ij}$

3) Adjoint $\rightarrow$ transpose of the cofactor element matrix

$$\begin{bmatrix} \begin{pmatrix} C_{11} \end{pmatrix} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^T$$

Adj(A) = $\begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{32} & C_{33} \end{bmatrix}$

Inverse of a matrix

$$2 \cdot \left(\frac{1}{2}\right) = 1$$

$$A \cdot A^{-1} = I$$

↳ Identity matrix

$$A^{-1} \neq \frac{1}{A}$$

$$\begin{bmatrix} 1 & 0 & & \\ 0 & 1 & & \\ 0 & 0 & \ddots & \\ & & & 1 \end{bmatrix}$$

square matrix
↳ scalar

Step 1. Calculate minor of the matrix

Step 2. Calculate cofactors

Step 3. Find adjoint for it

Step 4. Multiply $\text{adj}(A)$ by the reciprocal of $\det(A)$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj}(A)$$

↓

↳ determinant

Inverse

Scalar = $(ad - bc)$ ↳ $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \text{Area}$
3D $\rightarrow$ Volume

Linear combination



Transformation

change
in

